

Phase 1 Report - Inception

Jacob Woodard

Michelle Messer

Scarlett Shropshire

April 3, 2026

Understanding the Problem

As a small team of data scientists working for a major mobile app store, it is crucial to have a clear understanding of the problem we are trying to solve. Our team has been left with a base model that inputs text ratings, and predicts a score. However, many key stakeholders of the company have expressed frustrations with the current system, believing it to be highly inaccurate and too simple. This project is now in the hands of our team, and it is our job to improve it. The current backlog of reviews is over 100,000 across 500 different apps, and completing a model to satisfy the needs of such a large customer base is critical.

Assigning Major Roles

- **Scarlett Shropshire** – Conducting Research on Tasks, Maintaining Documentation Standards, Architecture Diagram
- **Michelle Messer** – ML Model Management, Vision Document, Github management, Connecting ML model to website
- **Jacob Woodard** – Maintaining Project Webpage, Connecting ML model to website, Github management, Vision Document

Although we may have designated roles, Michelle and Jacob share several overlapping responsibilities and will divide these tasks evenly to ensure balanced contributions. All members are expected to maintain detailed documentation to give to other members of the

team. This will ensure team members are all on the same page and are knowledgeable on every step of the process.

Though we may have designated roles, all members are expected to maintain detailed documentation to give to other members of the team. This will ensure team members are all on the same page and are knowledgeable on every step of the process.

Stakeholders and Personas

We can make plenty of assumptions of what key stakeholders want out of this product and make improvement decisions based on those assumptions. However, it's best to communicate with them directly to understand their exact concerns and desired outcomes. Due to the size of this company, there are a plethora of stakeholders that have an interest in this product. From app developers, app store managers, the end-users, platform owners, advertisers, and internal review teams, each stakeholder has different wants and needs when it comes to the use of this project. To get a better understanding of the key motivations and pain points of potential stakeholders, three personas were drafted to gain insight into specific stakeholders.

Morgan Chen: An App Store Moderator / Quality Manager. Morgan's primary job is to moderate the quality of apps across the app store, and she sees the scoring system as, a way to simplify her work process. The scoring system allows her to surface problems quickly instead of having to manually read reviews. However she believes that the current model gives unreliable and inconsistent scores, and that the system does not account for biased reviews in the data. She has little trust in the system that she uses daily to streamline her work process.

Maximilian Stein: Mid-Level Mobile App Developer. Maximilian manages performance features for his company, so it is crucial to have accurate and reliable feedback. He wants to utilize the scoring system to track negative reviews and the type of performance issues that are being mentioned. However, he is very frustrated that he has to manually check reviews due to the scoring system inaccurately flagging reviews with bugs mentioned in them as positive. Though Maximilian only uses the scoring system a couple times a week, he heavily relies on the accuracy of results to guide his workflow when developing his performance improvements.

Hailey Irving: App Advertiser for Growing Social Media App. Hailey is head of advertising for EchoSphere, a growing social media app. Hailey wishes to use the scoring system to find the best apps on which to spend her limited advertising budget to run ads on. She also wants to use the system to quickly find which apps to avoid or pull existing ads from. She cannot use the scoring system in its current state, and is frustrated that she has to

spend her limited time doing manual research for the best apps to advertise in. An improved scoring system would ideally help her save time and money.

Vision Document (Summary)

Now that we have a better idea of who our stakeholders are, their wants, and their frustrations with the system, we can set a concrete vision for the direction of the system. The stakeholders stressed that their biggest frustration with the system is its unreliable results, and its simple scoring system. The stakeholders rely on this system to streamline their workflow and cut down the time dedicated to research, so it's imperative to have an accurate system. The updated app scoring system will accept unstructured text as input and return a predicted rating for each review, maintaining a minimum accuracy of 80% with a target range of 85–90%. The system will also incorporate filtering functionality to categorize reviews into three bins: positive, neutral, and negative, while sorting reviews on a more granular 1–10 scoring scale. This will give greater complexity to the system, allowing the stakeholders to have more versatility in its utilization.

The updated system will introduce consistency and reliability, empowering stakeholders to make confident, data-driven decisions that ultimately benefit end users. For app store managers, the system provides clearer visibility into which apps are consistently underperforming and in need of further inspection, supporting decisions such as demotion or removal. For app developers, the complaint filtering and trend series features make it easier to pinpoint bugs and prioritize improvements within their apps. For external partners such as advertisers, the system helps identify the most reviewed and top-rated apps, making promotional decisions more targeted and informed. Overall, the updated system addresses the needs of all key stakeholders, delivering consistency, reliability, and efficiency across the board.

Functional Requirements (Summary)

- The system should filter reviews by category, display frequency of complaints, display ratings in ascending order, and detect spam reviews.
- The system should have a scoring system on a 1-10 scale, provide a confidence score on a 0-100% scale, and reflect the context of the review accurately.
- The system should flag apps with scores below 4.0 for inspection and set aside score with 10.0 for advertising stakeholders.

- The system should flag reviews that contain only a single word, reviews with negative ratings with no additional context, and display a warning indication when confidence scores drop below 60%.

Non-Functional Requirements (Summary)

- The system should have a minimum prediction accuracy of 80% and should flag at least 95% of single word reviews.
- The system should display confidence scores with $\pm 2\%$ precision.
- The system should filter/sort reviews with only 2 interactions, load under 5 seconds, support at least 300 concurrent users, support around 10,000 reviews a week, and retain trend score history for at least 6 months.
- The system should maintain detailed documentation so that a new developer can understand the pipeline within 2 hours.